

Part 2: Text Data Analysis

Task 1: Topic Modelling Analysis using LDA

Introduction

Today, the sheer amount of information, particularly in various unstructured formats like images and text, make it difficult for agencies and researchers to analyse and store these data in an efficient method. This project will highlight the methods used to analyse unstructured data in the form of texts.

This project utilises techniques like text mining and natural language processing in R to analyse the contents of 80 news articles, across two topics: sports, and science and technology. The news articles are collected from online news sites, mainly from The Malay Mail, but some articles are taken from South China Morning Post and The Star. Then, by transforming the raw text into a Document-Term Matrix (DTM), which has more structure, statistical models can be used to identify the patterns that are hidden within the text.

Methodology

The news articles are compiled into an excel file with two columns: title and content. An R script then reads this excel file and converts it to individual text files based on the articles. The text files are formatted as shown in figure 1 below.

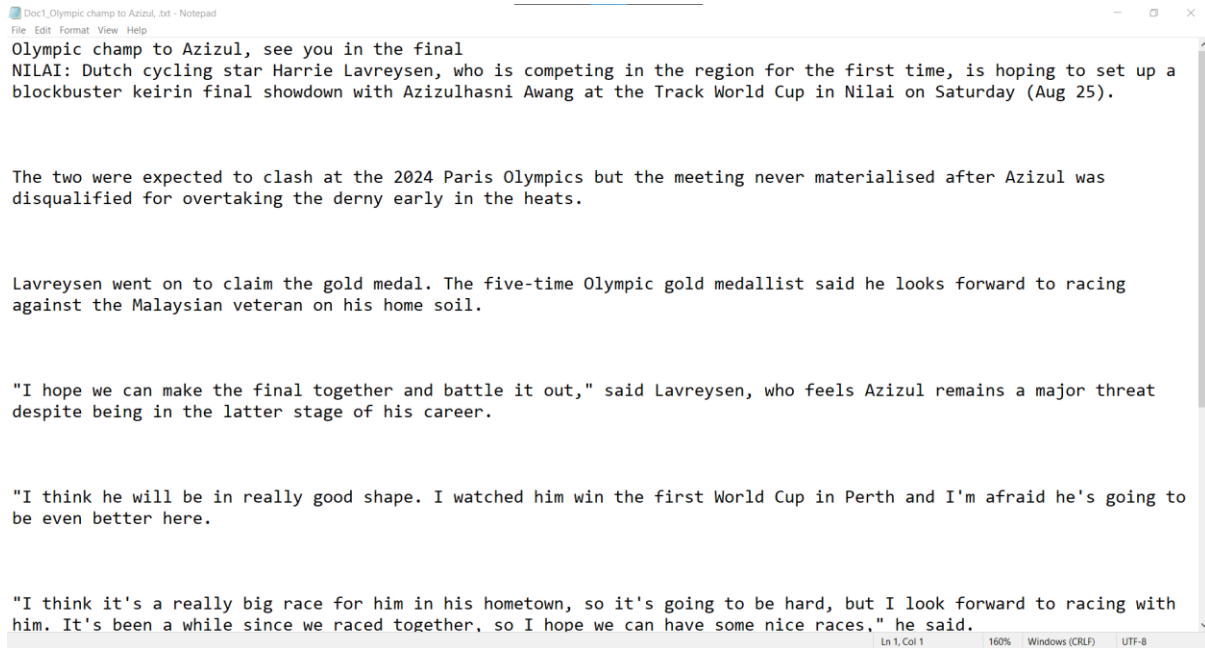


Figure 5: Contents of sample article in text file

The title of the article is appended to the first line of the text file, and the contents of the text file begin on the second line. The text files are named with an index and a title based on the shortened version of the title of the article. A sample of titles of the text files are shown in figure 2 below.

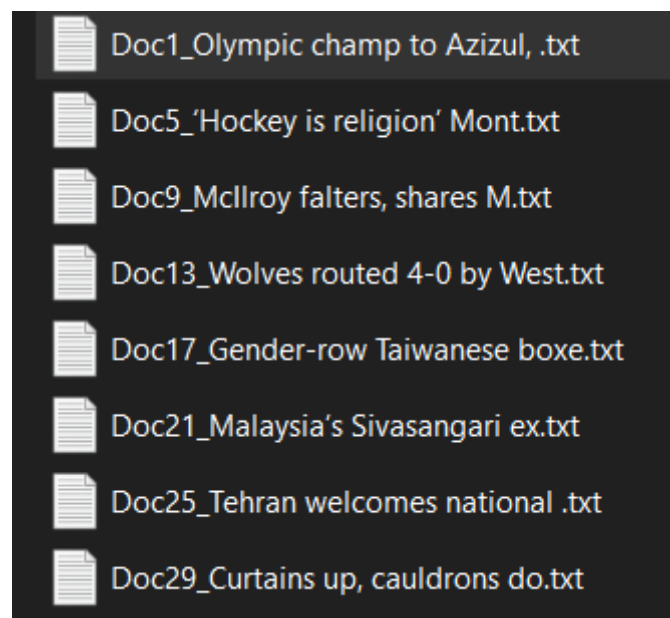


Figure 6: Sample of text file titles

Naming the files this way ensures that when visualising the results of the data analysis, the plots are not messy and can be viewed easily.

Once the individual text files are created, all these files are converted into a corpus. Afterwards, several pre-processing steps are taken to ensure that the data is as clean as possible.

The first step to pre-process the texts is to remove the title from line 1, location, and dates by collapsing multiple lines into one singular string. Then, contractions like “n’t” are replaced to “not”, and contractions like “’s, ’re, ’ve, ’ll, ’d, ’m” are removed. Next, the author’s name at the end of the article is removed, which is preceded by an emdash. The texts are then converted to lowercase, hyphens and punctuations are removed and the texts are lemmatized. A custom list of words ("say", "much", "will", "can", "time", "good") and English stopwords are removed to get a better distinction between topics. Lastly, whitespace between words is removed to conclude the pre-processing step. The corpus is then converted to a Document-Term Matrix.

Once this is done, Latent Dirichlet Allocation (LDA) is used to perform topic modelling on the articles.

Results and Discussion

From the LDA analysis, the per-topic-per-word probabilities, beta spread, and per-document topic probabilities can be obtained.

Per-topic-per-word probabilities

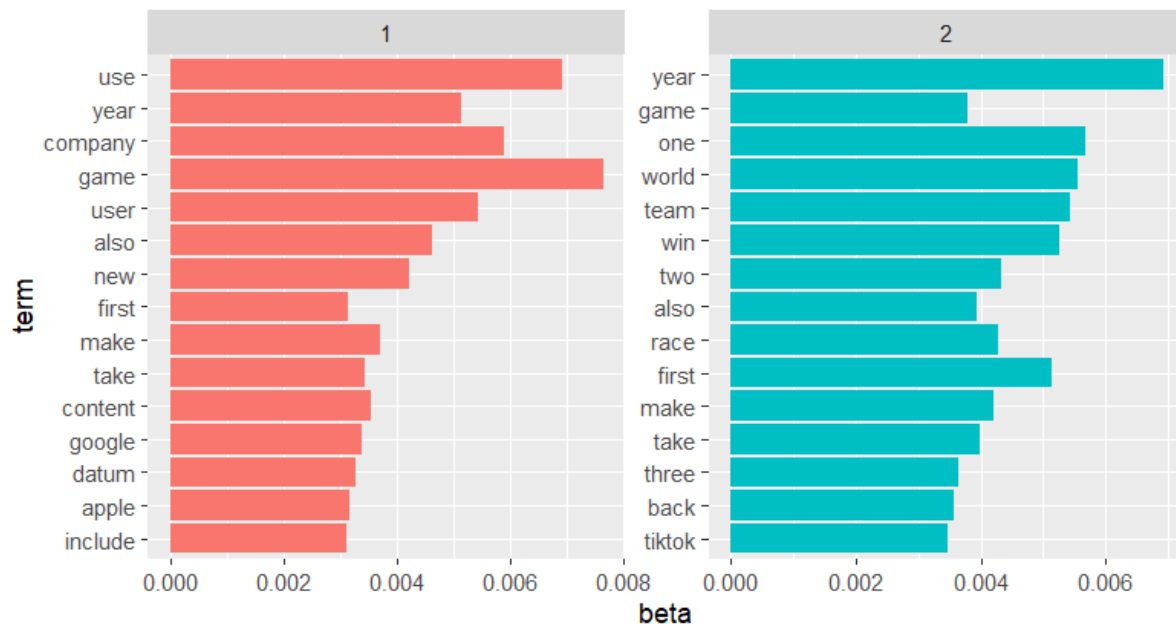


Figure 7: Probabilities of terms for each topic

The illustration in figure 7 above shows the probabilities of terms for each topic. From the probabilities of these terms, an inference can be made that topic 1 is most likely to be science and technology, while topic 2 is more likely to be sports. This chart helps us understand the two topics that were inferred by the model from the article text data. It is also important to note that some terms overlap between topics, such as the word “make”, which is an advantage of topic modelling over clustering methods as topics could have some overlap in words.

Beta spread

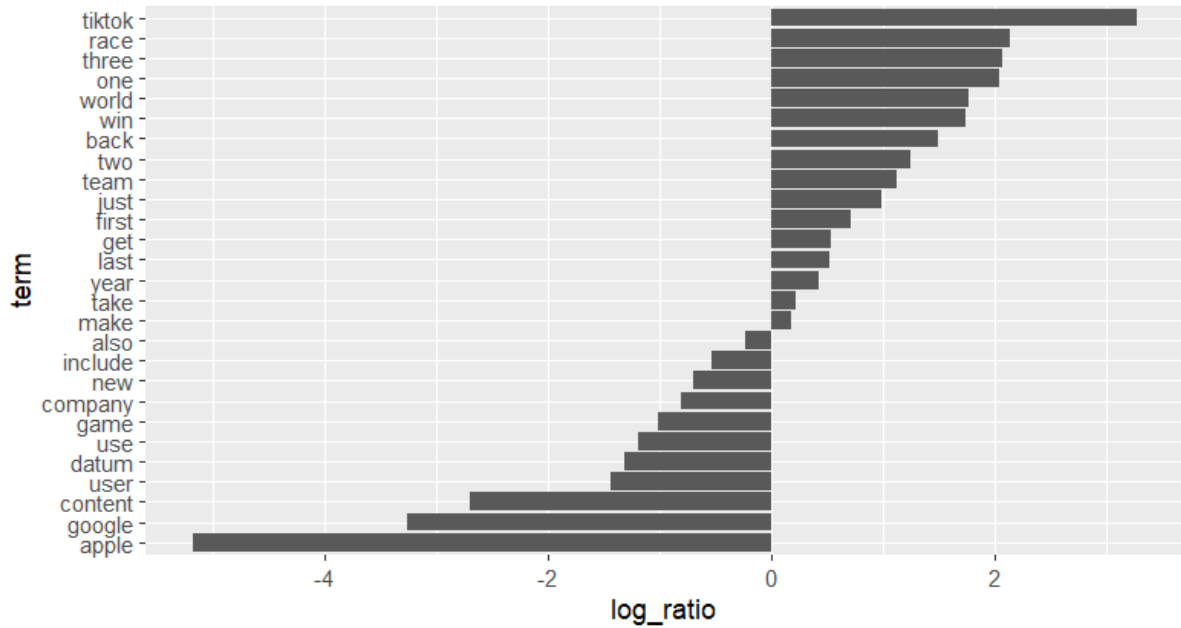


Figure 8: Beta spread for tems

Figure 8 shows the beta spread for terms present in the articles. The spread can be estimated based on the logarithmic ratio between the probabilities of each term in each topic, as shown in the expression below.

$$\log \left(\frac{\beta_2}{\beta_1} \right)$$

Using ratios makes the difference between the likelihood of terms appearing in each topic symmetrical. If β_2 is greater than β_1 , then the log ratio will be positive while the opposite is true if the log ratio is negative. As shown in the chart, the words “tiktok” and “race” have an extremely large ratio which strongly indicates that these two words belong to topic 2, which is sports, while words like “apple” and “google”, which show a negative ratio, indicates that these words most likely belong to topic 1, which is science and technology.

Per-document-per-topic probabilities

Table 2: Sample of per-document-per-topic probabilities

document	topic	gamma
Doc80_She married an AI persona.txt	2	0.999944
Doc70_From robot arms to REDMI .txt	1	0.999917
Doc60_From ChatGPT to robot but.txt	1	0.999903
Doc41_Level playing field Tech .txt	2	0.999897
Doc19_You're being watched Japa.txt	2	0.999896
Doc76_TikTok's algorithm is the.txt	2	0.999894
Doc14_Real Madrid held to 1-1 d.txt	1	0.999892
Doc52_The Grok scandal explaine.txt	2	0.999892
Doc68_Tiny tech, big AI power 2.txt	1	0.999889
Doc66_Trade wars and chip short.txt	1	0.999888
Doc73_Resurrected without conse.txt	2	0.999886
Doc78_Explained How TikTok keep.txt	2	0.999886
Doc7_Arsenal suffer major Prem.txt	1	0.999885
Doc9_McIlroy falters, shares M.txt	1	0.999885
Doc10_Pogacar falls short at Pa.txt	1	0.999883
Doc62_Listen up earbuds may b.txt	2	0.999881
Doc49_ChatGPT is getting ads .txt	2	0.999879
Doc54_AI pendants and wearables.txt	1	0.999879
Doc64_Big screens, bigger ads S.txt	1	0.999879

Table 2 shows the per-document-per-topic probabilities for a sample of 20 documents in descending gamma order. This table shows the likelihood a document belongs to a certain topic. As seen from the table, most science and technology articles are correctly predicted to be in topic 2, however, some documents that are supposed to belong to topic 1 (science and technology), like document 80 are incorrectly predicted to belong to topic 2, which is sports.

Confusion Matrix

A confusion matrix was plotted using the maximum per-document-per-topic probabilities (γ) to explain the results obtained by the LDA analysis.

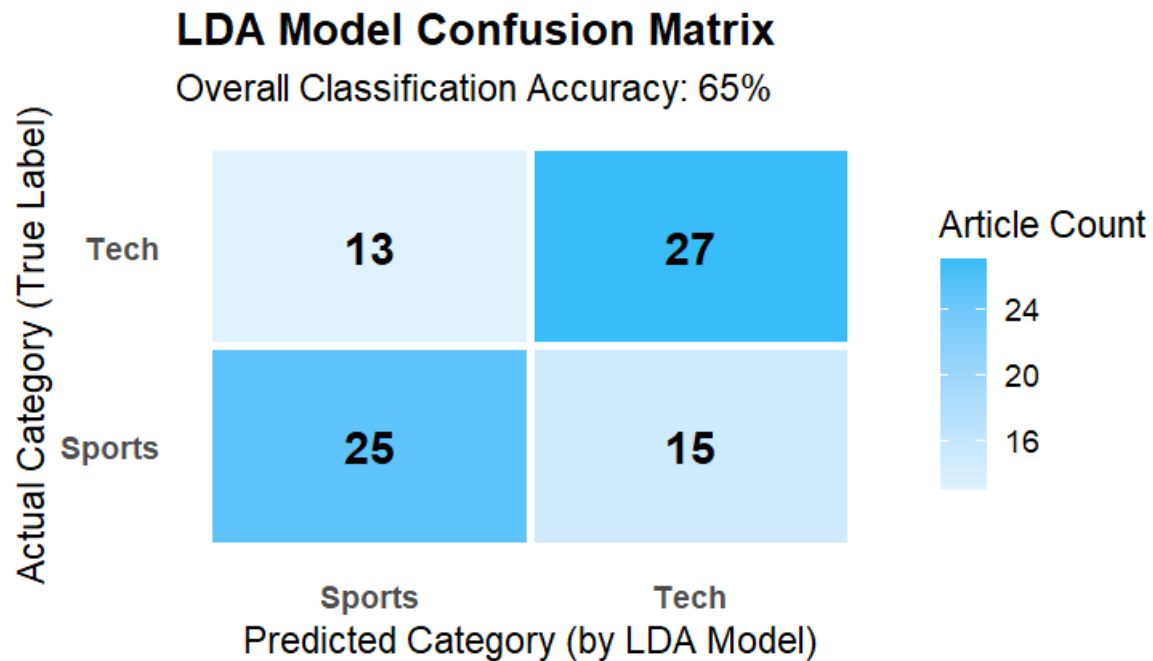


Figure 9: Confusion Matrix

From the confusion matrix in figure 9, the LDA achieved a moderate classification accuracy of 65%. Out of 80 total documents, the model correctly mapped 52 documents. For the sports category, 25 articles were correctly identified as belonging to the sports category, while 15 articles were incorrectly identified as belonging to the science and technology category, which gives a classification accuracy of 62.5% for the sports category. For the science and technology category, 27 were classified correctly, while 13 articles were incorrectly classified as belonging to the sports category, giving a classification accuracy of 67.5%, which is slightly better than the accuracy for the sports category.

WordCloud

From the document-term matrix, a wordcloud can also be produced to visualize the frequent terms that appear in the articles. The 50 most common words are filtered and shown on the wordcloud.



Figure 10: Wordcloud for news articles

From the wordcloud in figure 10, the most common terms are “year”, “game”, “make”, “team”, “company”, and “use”. These words are commonly repeated in both articles, since they are “generic” words. LDA classifies document topics based on frequencies, so these words act as “noise” and reduces the distinction between the sports and science and technology topics.

There is also overlap between specific words like “game” which can belong to either sports topics or science and technology topics. This can “confuse” the model, making it incorrectly classify articles.

Model Improvement

The LDA model's classification accuracy can be improved by using a method called "Gibbs Sampling". The Gibbs Sampling algorithm is a Monte Carlo Markov-chain algorithm, powerful technique in statistical inference, and a method of generating a sample from a joint distribution when only conditional distributions of each variable can be efficiently computed and is widely used in research (Jelodar et al., 2019)(Porteus et al., 2008). By default, the LDA function uses the VEM (Variational Expectation-Maximization) method (Grün et al., 2026).

To achieve this in R, we can specify the method argument to "Gibbs" and provide the number of iterations for the model to run. The code snippet is shown below.

```
#Perform LDA analysis
article_LDA <- LDA(article_dtm, k=2, method="Gibbs",
                    control = list(seed = 1234, iter = 2000, burnin = 500))
```

Figure 11: Code snippet for Gibbs Sampling method implementation in R

By doing this, we increase the classification accuracy to 98.75%, as shown in the confusion matrix below.

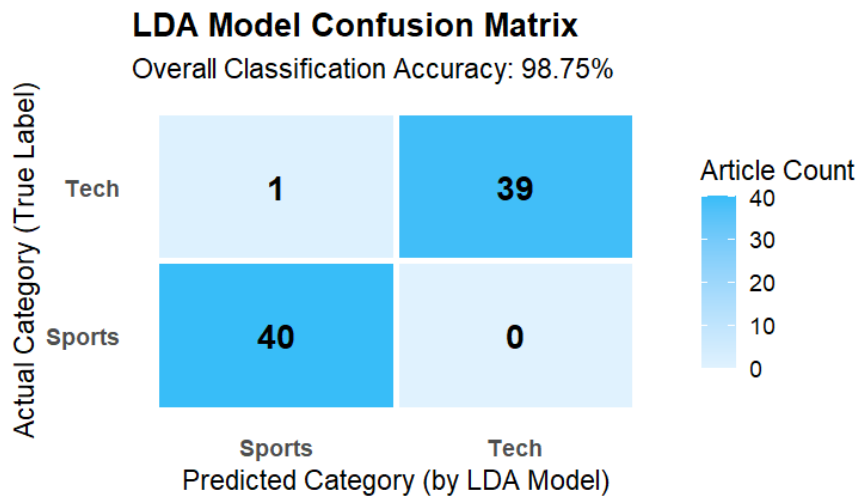


Figure 12: Confusion matrix using Gibbs Sampling

Comparing to the previous confusion matrix in figure 9, this time, the model can correctly classify all sports articles, giving a classification accuracy of 100%. While only one article from the science and technology article is incorrectly classified as a sports article, which gives the classification accuracy of 97.5% for the science and technology topic. This shows that the model can get a better distinction between topics using the Gibbs sampling method.

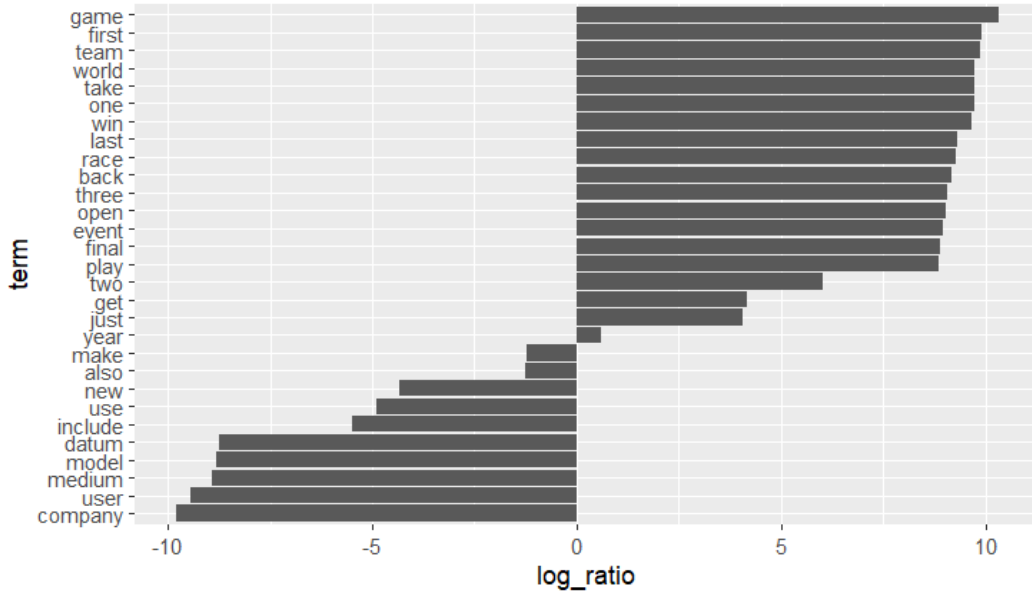


Figure 13: Beta spread obtained using Gibbs Sampling

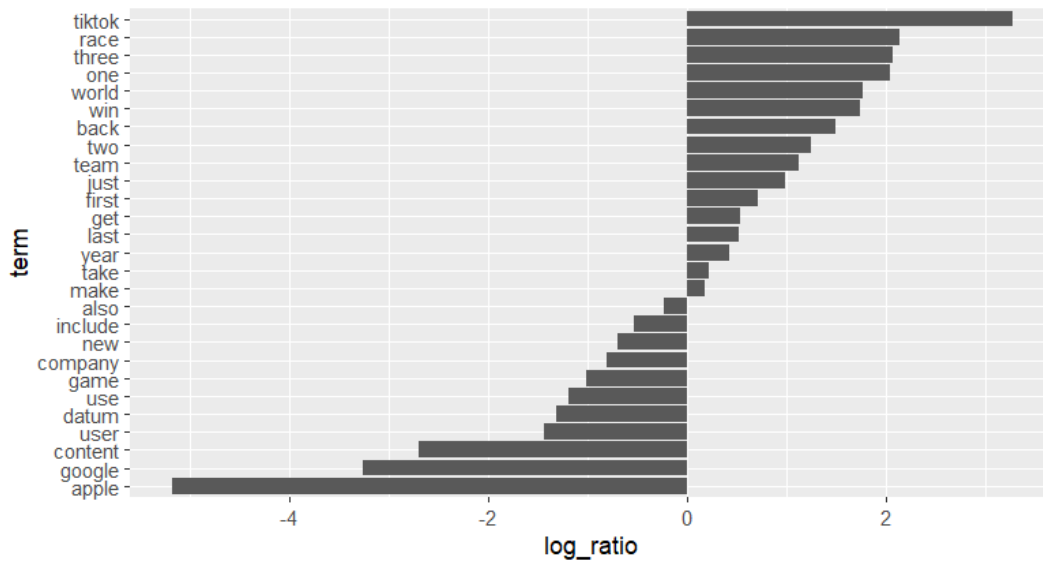


Figure 14: Beta spread obtained using default VEM method

Comparing the beta spread plot in figure 13 to the beta spread plot in figure 14, the model can better understand the distinction between words that appear in each topic and correctly classify them. The key difference is in the log ratio. In the plot using the Gibbs sampling method, the log ratio is between -10 to +10, whereas without using the Gibbs sampling method, the log ratio is only roughly between -5 to +3. A higher log ratio range indicates that there is a higher chance of a specific word to belong to one topic over the other. We can get the inverse of the log ratio function used, which we defined previously, to get how likely a term will appear in one topic over the other, which results in 2^n . For a log ratio of +10, the word “game” has a $2^{10} = 1024$ times a higher chance to appear in topic 2, which is sports.

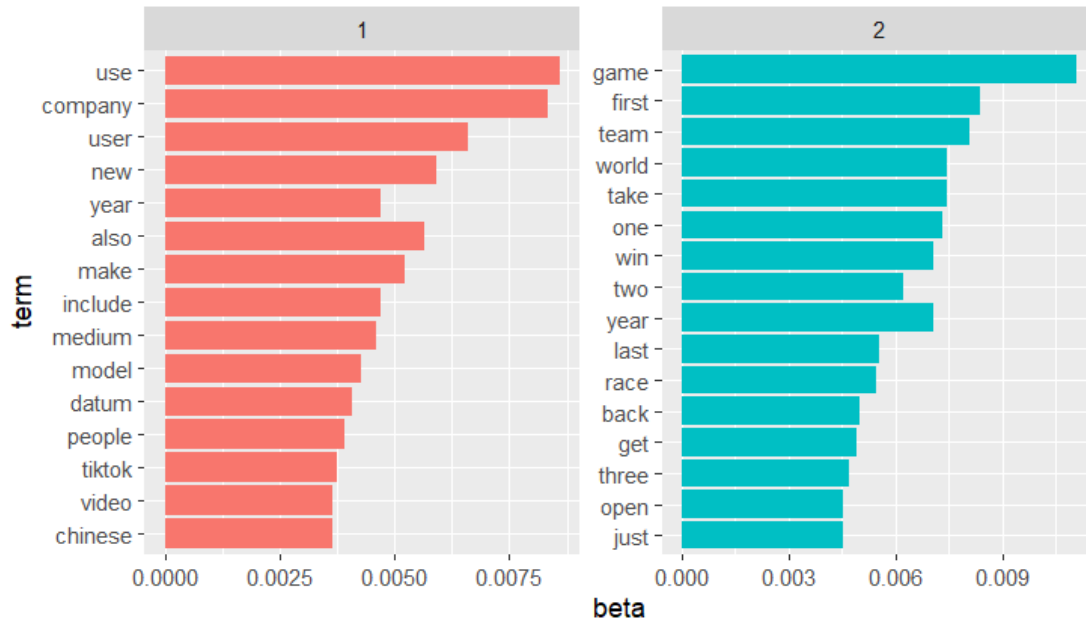


Figure 15: Topic-term probabilities using Gibbs Sampling

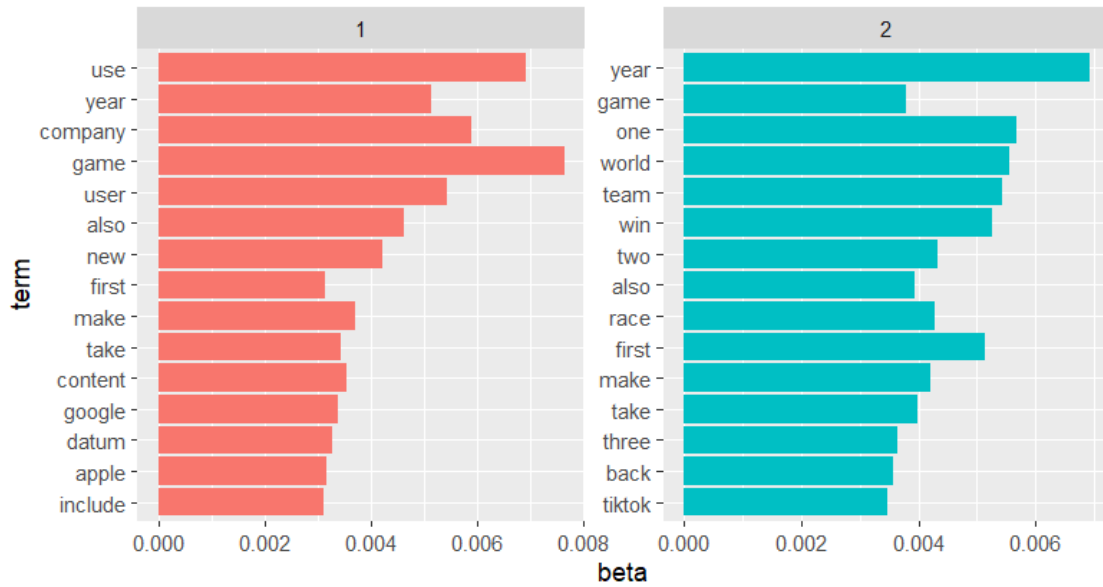


Figure 16: Topic-term probabilities using default VEM method

For the term-topic probability plot, the Gibbs sampling method classifies the terms more accurately. For example, the word “tiktok” which previously was classified into the second topic (sports), as seen in figure 16, is now classified into the first topic (science and technology), as seen in figure 15. Certain overlaps have also been removed, like the word “game” which previously appeared in both topics. Using the Gibbs sampling method, the word “game” is now predicted to be in topic 2, with a much higher probability, indicating that the word “game” strongly belongs to topic 2.

Conclusion

In conclusion, LDA is a useful tool to identify and quantify underlying topics in raw text documents. The performance of LDA depends on hyperparameter tuning and algorithm selection. As seen from the results, the default VEM approach yielded a moderate classification accuracy of 65%. Further analysis through wordcloud and beta spread plots showed that the reason why the default model underperforms is due to “thematic bleeding” where the model cannot clearly distinguish the two topics. Generic words like “team”, “game”, and “year” which are potentially present in articles of both topics confuse the model, and thus, limits its overall classification accuracy.

By implementing Gibbs Sampling, we increased the classification accuracy to 98.75%. In essence, the Gibbs Sampling method “re-engineered” how the model works, providing an increased log ratio of -10 to +10, which allows the model to better separate and categorize terms according to the underlying themes contained in the articles.